

COMPARISON OF MACHINE LEARNING ALGORITHMS AND THEIR ENSEMBLES FOR BOTNET DETECTION

by
Songhui Ryu

A Thesis

*Submitted to the Faculty of Purdue University
In Partial Fulfillment of the Requirements for the Degree of*

Master of Science



Department of Computer and Information Technology

West Lafayette, Indiana

May 2018

ProQuest Number: 10808417

All rights reserved

INFORMATION TO ALL USERS

The quality of this reproduction is dependent upon the quality of the copy submitted.

In the unlikely event that the author did not send a complete manuscript and there are missing pages, these will be noted. Also, if material had to be removed, a note will indicate the deletion.



ProQuest 10808417

Published by ProQuest LLC (2018). Copyright of the Dissertation is held by the Author.

All rights reserved.

This work is protected against unauthorized copying under Title 17, United States Code
Microform Edition © ProQuest LLC.

ProQuest LLC.
789 East Eisenhower Parkway
P.O. Box 1346
Ann Arbor, MI 48106 – 1346

THE PURDUE UNIVERSITY GRADUATE SCHOOL
STATEMENT OF COMMITTEE APPROVAL

Dr. John Springer, Chair

Department of Computer and Information Technology

Dr. Baijian Yang

Department of Computer and Information Technology

Dr. Eric Matson

Department of Computer and Information Technology

Approved by:

Prof. Eric T. Matson

Head of the Graduate Program

ACKNOWLEDGMENTS

I wish to gratefully acknowledge my thesis advisor, Dr. John Springer for his insightful comments and guidance. He was always open to help me and willing to solve my doubts whenever I struggled with my research question or writing. He consistently allowed this paper to be my own work, but encouraged me to proceed in the right the direction whenever he thought I needed it. I would also love to acknowledge my thesis committee members, Dr. Eric Matson and Dr. Baijian Yang. They helped me to constantly stick to my research providing motivation and opportunities to attend conferences to meet other researchers in the same area. Also, I would like to thank to my ITaP coworkers, Alex Younts and Preston Smith who gave me the opportunity to work as a research assistant. By working with them, I was able to kept myself motivated and learning the computing environment for big data analysis. Finally, I appreciate all the support and encouragement that my family and my friends have been giving to me. Their trust in me made my thesis and achievement possible. Thank you.

TABLE OF CONTENTS

LIST OF TABLES	vi
LIST OF FIGURES	vii
LIST OF ABBREVIATIONS	viii
GLOSSARY	ix
ABSTRACT	x
CHAPTER 1. INTRODUCTION	1
1.1 Research Question	1
1.2 Scope	1
1.3 Significance	2
1.4 Assumptions	3
1.5 Limitations	4
1.6 Delimitation	4
1.7 Summary	5
CHAPTER 2. REVIEW OF LITERATURE	6
2.1 Botnet	6
2.2 Machine Learning for Classification	7
2.2.1 Naive Bayes	8
2.2.2 Artificial Neural Networks (Multi-layer Perceptron)	9
2.2.3 Decision Tree	9
2.3 Ensemble Methods	10
2.3.1 Voting	10
2.3.2 Bagging	11
2.3.3 Boosting	11
2.3.4 Random forest	12
2.4 Related Works	13
2.4.1 BotSniffer	13
2.4.2 BotMiner	14

2.4.3 BotHunter	15
2.5 Summary	15
CHAPTER 3. METHODOLOGY	16
3.1 Research Overview	16
3.2 Dataset	17
3.3 Accuracy Metrics	19
3.3.1 F1 score	19
3.3.2 Matthews Correlation Coefficient(MCC)	20
3.4 Machine Learning Library	21
3.5 Data preparation and evaluation	22
3.6 Deliverables	23
3.7 Summary	24
CHAPTER 4. RESULTS	25
4.1 Results	25
4.2 Summary	30
CHAPTER 5. CONCLUSION	31
5.1 Discussion	31
5.2 Comparison with the benchmarks	32
5.3 Future work	33
5.4 Summary	34
REFERENCES	36
APPENDIX A. DATA PREPARATION	42

LIST OF TABLES

3.1	Data features in the CTU-13 dataset (Garcia, Grill, Stiborek, & Zunino, 2014)	23
4.1	Time consumed for model training (sec)	25
4.2	Evaluation result	27

PREVIEW

LIST OF FIGURES

2.1	A centralized botnet architecture.	7
2.2	A decentralized P2P botnet architecture.	7
3.1	Overview of the experiment.	17
4.1	Experiment results	26
4.2	Time and MCC evaluation against S4	28
4.3	Time and MCC evaluation against S10	29
4.4	Time and MCC evaluation against S11	29

PREVIEW

LIST OF ABBREVIATIONS

AdaBoost	Adaptive Boosting
DT	Decision Tree
FN	False Negative
FP	False Positive
GNB	Gaussian Naive Bayes
IDS	Intrusion Detection System
MCC	Matthews Correlation Coefficient
ML	Machine Learning
NN	Neural Networks
TN	True Negative
TP	True Positive

GLOSSARY

Botnet – A botnet is a network of compromised devices used by a botnet owner to perform various malicious tasks. The devices – often personal computers – called “Bots” are under the control of a human “Botmaster.”

Ensemble method - An ensemble method is an approach that makes a set of classifiers into an ensemble by combining the prediction from each classifier possibly with weights. It is regarded as one of the methods for improving the accuracy.

Machine learning algorithm - A machine learning algorithm refers to an algorithm that is used to generate a statistical model from input data for various purposes from new incoming data. In practice, the former phase is called training, and the latter is called testing.

ABSTRACT

Author: Ryu, Songhui. M.S.

Institution: Purdue University

Degree Received: May 2018

Title: Comparison of Machine Learning Algorithms and Their Ensembles for Botnet Detection

Major Professor: John Springer

A Botnet is a network of compromised devices controlled by a botmaster often for nefarious purposes. Analyzing network traffic to detect Botnet traffic has historically been an effective approach for systems monitoring for network intrusion. Although such system have been applying various machine learning techniques, little investigation into a comparison of machine algorithms and their ensembles has been undertaken. In this study, three popular classification machine learning algorithms – Naive Bayes, Decision tree, and Neural network – as well as the ensemble methods known to strengthen said classifiers are evaluated for enhanced results related to Botnet detection. This evaluation is conducted with the CTU-13 public dataset, measuring the training time and accuracy scores of each classifier.